Mobility protocol — upper body day (hEDS / HSD safe)

Pre: neuromuscular activation · Post: active recovery + CARs · NO passive stretching · matched to your routine

Target area

Shoulder / rotator cuff · Chest · Back / lats · Biceps / triceps · Wrists / t-spine

⚠ Why no passive stretching — hEDS / HSD principle

Your connective tissue already has excessive laxity. Passive stretching bypasses muscles and loads ligaments/joint capsules instead → more instability. Every drill below keeps muscles ACTIVE throughout the entire range so joints stay protected while tissue quality improves.

PRE-WORKOUT — Activation + Joint Priming (~14 min · before warmup sets)

Goal: wake up stabilizers · build joint awareness · prime neuromuscular control · muscles ON throughout

P1 Shoulder CARs (controlled articular rotations) — 3 each direction × each arm ~30s per circle
- Entire shoulder capsule · rotator cuff · scapular stabilizers
- Slow full-circle rotation — create maximum tension through entire body
- Fist clenched, brace core, irradiate tension — "move the arm through wet concrete"

P2 Wrist CARs (fist closed, forearm braced with opposite hand) — 5 each direction × each wrist ~15s per circle
- Wrist capsule · forearm flexors + extensors
- Lock forearm still with opposite hand — only wrist moves
- Priority joint for you — slow circles, feel for any catching/clicking and note it

P3 Scapular push-up isometric holds (protract + hold at top) — 3 × 10s hold From knees if needed
- Serratus anterior · lower trap · scapular stabilizers
- Plank position → push shoulder blades apart maximally → HOLD
- Teaches scapulae to "stick" to ribcage — critical before pressing movements

P4 Band isometric external rotation — 3 angles — 8s hold × 3 angles × 2 rounds · each arm Light band
- Infraspinatus · teres minor · posterior cuff
- Elbow pinned to side → hold band at 3 positions: 0°, 45°, 90° external rotation
- Builds active control at the ranges where your shoulders are most vulnerable

P5 Cat-cow with 3s isometric pause at each end — 2 × 6 reps 3s hold each end
- Thoracic spine · erector spinae · serratus anterior · deep core
- Pause in cat: push floor away hard (serratus activation)
- Pause in cow: squeeze between shoulder blades (rhomboid activation)

P6 Wrist isometric loading — 4 directions (tabletop position) — 5s hold × 4 dirs × 2 rounds
- Wrist flexors · extensors · radial + ulnar deviators
- Hands flat on floor → lean gently into: forward / back / left / right → hold
- Teaches your wrists to handle load in all directions before gripping heavy

P7 Active pec opener — band pull-apart at chest height + hold — 3 × 8 reps 5s hold each rep Light band
- Pec major · pec minor · rhomboids · posterior delt
- Pull band apart until mild stretch in chest → HOLD 5s with rhomboids squeezing
- This "opens" the chest actively — muscles control the range, not passive joint laxity

POST-WORKOUT — Active Recovery + CARs (~16 min · after last set)

Goal: down-regulate nervous system · restore joint awareness · reduce post-workout clicking · muscles engaged throughout

S1 Shoulder CARs — slow recovery tempo — 2 each direction × each arm ~45s per circle
- Full shoulder capsule assessment + recovery
- Same as pre-workout but slower (~45s per circle) — scan for new clicking/catching
- Compare feel to pre-workout — if worse, note it for next session planning

S2 Slow eccentric push-up — 8s lowering (from knees OK) — 3 reps 8s eccentric each From knees = fine
- Pec major · anterior delt · triceps · serratus anterior
- Lower yourself over 8 seconds — chest muscles lengthen UNDER LOAD
- This is your "chest stretch" — eccentrics lengthen muscle tissue safely with joints protected

S3 Active hang — bent arm (feet on floor for partial weight) — 2 × 15s hold Feet on floor Elbows soft
- Lats · teres major · rhomboids · grip + forearm
- Hang from bar with elbows slightly bent, feet on floor taking 30–50% bodyweight
- Keep shoulders ACTIVELY packed down — never relax into a dead hang
- This replaces passive lat stretching — decompresses spine while lats stay engaged

S4 Isometric biceps lengthened hold (arm extended, palm on wall, push) — 2 × 15s × each arm ~30% effort
- Biceps brachii · brachialis · anterior forearm
- Arm at ~160° (NOT locked straight) → push palm into wall with ~30% effort
- Muscle is active in lengthened position — gets the "stretch" effect while protecting elbow

S5 Isometric triceps overhead hold (towel or band behind back) — 2 × 12s × each arm Towel / light band
- Long head of triceps · posterior shoulder
- Top hand pulls towel up, bottom hand resists → triceps lengthened under tension
- Hold at mild "stretch" position — the 30% push keeps joint centrated

S6 Quadruped t-spine rotation with active reach (hand behind head) — 2 × 5 each side 3s push at top
- Thoracic rotation · obliques · rhomboids · mid-trap
- Hand behind head → rotate open → at end range PUSH elbow toward ceiling 3s
- The push at end range makes this active — replaces passive thread-the-needle

S7 Wrist CARs (recovery tempo) + 4-direction isometric press — 3 CARs each way + 5s × 4 directions × each wrist
- Wrist capsule · forearm flexors + extensors
- Slow CARs first → then press palm into opposite hand: flex / extend / radial / ulnar
- Compare clicking to pre-workout — this is your wrist health check

S8 Prone Y-T-W isometric holds (face down on bench or floor) — 2 rounds 8s each letter Bodyweight only
- Lower trap · mid-trap · rhomboids · posterior cuff
- Thumbs up → lift arms into Y position 8s → T position 8s → W position 8s
- Low intensity scapular stability — "tucks" your shoulder blades back after all the volume

S9 90/90 diaphragmatic breathing (feet on bench, knees + hips at 90°) — 5 breath cycles 4s in / 8s out ~60s total
- Diaphragm · deep core · ribcage positioning · nervous system downregulation
- Inhale 4s through nose (ribs expand laterally) → exhale 8s through mouth (ribs drop)
- Resets ribcage position — especially important for hEDS rib subluxation tendency

Breathing cue for isometric holds: inhale before the hold → slow exhale DURING the hold

Exhaling under tension helps maintain core pressure without breath-holding spikes

Quick reference — muscle ↔ drill map (replaces passive stretching)

Shoulder capsule / rotator cuff	Pre: Shoulder CARs, Isometric ext. rotation · Post: Slow CARs, Y-T-W holds
Pec major / pec minor	Pre: Active band pec opener · Post: Eccentric push-up (8s lowering)
Lats / rhomboids / mid-trap	Pre: Cat-cow with iso pauses · Post: Active hang, Y-T-W holds
Biceps	Pre: (warmed by CARs) · Post: Isometric lengthened wall hold
Triceps	Pre: (warmed by scap push-ups) · Post: Isometric overhead towel hold
Thoracic spine	Pre: Cat-cow with iso pauses · Post: Quadruped t-spine rotation with push
Wrists	Pre: Wrist CARs + 4-dir iso loading · Post: Slow CARs + 4-dir iso press
Serratus anterior	Pre: Scapular push-up holds · Post: Y-T-W holds, 90/90 breathing
Ribcage / core stability	Pre: Cat-cow pauses · Post: 90/90 diaphragmatic breathing (rib reset)

Session self-check — track these over time

- Shoulder CARs: is the clicking better/worse/same compared to pre-workout?
- Wrist CARs: any new catching or pain points? Compare to pre-workout
- Overall joint stability: do joints feel more or less stable than when you walked in?

hEDS / HSD mobility rules — every session

- NEVER passively stretch — every "stretch" must have active muscle engagement
- Isometric holds: 20–30% effort only — goal is neuromuscular control, not strength
- CARs: if a portion of the circle catches or clicks, slow down there — don't force through
- No dead hangs — always keep shoulders actively packed, feet assisting
- If any joint feels MORE unstable after a drill — remove it next session and tell your physio

Session timeline

Activation + priming (14 min) → Warmup sets A (3 min) → Working sets B–E (27 min) → Active recovery + CARs (16 min)

Total session with mobility protocol: ~82 min

Collagen + Vit C 30–60 min before · Protein + Vit C within 60 min after